

# S2N XSS Security Report

Generated at: 2026-05-13T02:42:20.401770Z

## 1. Executive Summary

This report was generated from an actual S2N scan result and describes confirmed XSS findings.

|                    |                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------|
| Primary Finding    | Cross-Site Scripting Detected                                                                    |
| Primary Severity   | HIGH                                                                                             |
| Primary Confidence | FIRM                                                                                             |
| Target URL         | http://host.docker.internal:4280/vulnerabilities/xss_r/?name=%3Cscript%3Ealert(1)%3C%2Fscript%3E |
| Affected Parameter | name                                                                                             |
| Primary Payload    | <script>alert(1)</script>                                                                        |
| Total Findings     | 1                                                                                                |
| Severity Counts    | {'HIGH': 1}                                                                                      |
| Total Requests     | 1                                                                                                |
| Source Scanner     | s2n                                                                                              |
| Source Plugin      | xss                                                                                              |

The finding is confirmed because the scanner observed unsafe payload handling through HTTP reflection or browser execution evidence.

## 2. All Findings Summary

---

The S2N scan detected the following XSS finding records. The following sections summarize each finding in a vertical detail format and then analyze the primary finding.

### 2.1 Finding 1

|              |                                  |
|--------------|----------------------------------|
| ID           | xss-1                            |
| Severity     | HIGH                             |
| Confidence   | FIRM                             |
| Target Path  | /vulnerabilities/xss_r/?name=... |
| Parameter    | name                             |
| Payload      | <script>alert(1)</script>        |
| Verification | payload_reflected                |
| Reflected    | true                             |

### 3. Representative Finding Detail

---

|                     |                                  |
|---------------------|----------------------------------|
| ID                  | xss-1                            |
| Title               | Cross-Site Scripting Detected    |
| Vulnerability Type  | XSS                              |
| XSS Type            | reflected                        |
| CWE                 | CWE-79                           |
| OWASP Category      | Injection / Cross-Site Scripting |
| HTTP Method         | GET                              |
| Parameter           | name                             |
| Injection Context   | url_parameter                    |
| Reflection Detected | true                             |
| Reflected Value     | <script>alert(1)</script>        |
| Verification        | payload_reflected                |
| Risk Score          | 85                               |
| Likelihood          | High                             |
| Impact              | High                             |

#### Description

User-controlled input was reflected or executed in an unsafe browser/HTML context. This may allow JavaScript execution in a victim browser.

## 4. Evidence

---

### Raw Evidence

```
XSS evidence confirmed by scanner; payload=<script>alert(1)</script>;  
verification=payload_reflected; status_code=200; method=GET; parameter=name;  
discovery_mode=direct_url_validation;
```

### Response Snippet

```
XSS evidence confirmed by scanner; payload=<script>alert(1)</script>;  
verification=payload_reflected; status_code=200; method=GET; parameter=name;  
discovery_mode=direct_url_validation;
```

## 5. XSSAgent Judgement

---

|                 |                                    |
|-----------------|------------------------------------|
| Task            | xss_reflection_confirmation        |
| Should Run      | true                               |
| Context Known   | true                               |
| Selected Plugin | -                                  |
| Severity        | HIGH                               |
| Confidence      | 97                                 |
| Fallback        | xss_stored_confirmation            |
| Reason          | XSS evidence confirmed by scanner. |

## 6. Risk & Remediation

---

### Risk

|                 |                                                                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Severity        | HIGH                                                                                                                                           |
| Risk Score      | 85                                                                                                                                             |
| Likelihood      | High                                                                                                                                           |
| Impact          | High                                                                                                                                           |
| Business Impact | An attacker may execute arbitrary JavaScript in a victim browser, steal session data, perform actions as the user, or manipulate page content. |

### Scoring Basis

- Severity is high because browser-executable JavaScript was confirmed by scanner evidence.
- Confidence is firm because the payload and execution/reflection evidence were verified by the scanner.
- Likelihood is high when user-controlled input is returned without context-aware output encoding.
- Impact is high because reflected script execution can lead to session theft, user action abuse, or page manipulation.

### Recommended Remediation

1. Apply context-aware output encoding before inserting user-controlled input into HTML.
2. Validate and constrain user input on the server side.
3. Avoid directly reflecting raw request parameters into responses.
4. Use safe DOM APIs such as `textContent` instead of `innerHTML` where possible.
5. Add Content Security Policy as a defense-in-depth control.
6. Re-test the affected parameter with the same payload after remediation.

# 7. Internal Knowledge Retrieved by ChromaDB

## 7.1 Recommended Fix Summary

|                 |                                         |
|-----------------|-----------------------------------------|
| Source          | docs/security_guides/xss_remediation.md |
| Retriever       | chroma                                  |
| Retrieval Score | 0.8905                                  |

Recommended Fix

- Apply context-aware output encoding.
- Escape HTML body output using HTML entity encoding.
- Escape HTML attribute values.
- Avoid inserting user input directly into JavaScript context.
- Prefer safe DOM APIs such as `textContent` instead of `innerHTML`.
- Validate and normalize input on the server side.
- Use Content Security Policy as a defense-in-depth control.

## 7.2 HTML Body Context Guidance

|                 |                                         |
|-----------------|-----------------------------------------|
| Source          | docs/security_guides/xss_remediation.md |
| Retriever       | chroma                                  |
| Retrieval Score | 0.8852                                  |

HTML Body Context

When untrusted input is rendered inside the HTML body, apply HTML entity encoding before rendering the value. For example, `<script>alert(1)</script>` should be rendered as `&lt;script&gt;alert(1)&lt;/script&gt;`.

## 7.3 Reflected XSS Explanation

|                 |                                         |
|-----------------|-----------------------------------------|
| Source          | docs/security_guides/xss_remediation.md |
| Retriever       | chroma                                  |
| Retrieval Score | 0.8755                                  |

Reflected XSS

Reflected XSS occurs when user-controlled input is included in an HTTP response without proper output encoding. If the browser interprets the reflected input as HTML or JavaScript, an attacker may execute script in the victim's browser context.

## 8. Official Reference Mapping

---

### 8.1 OWASP - OWASP XSS Prevention Cheat Sheet

|             |                                                                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source      | <a href="https://cheatsheetseries.owasp.org/cheatsheets/Cross_Site_Scripting_Prevention_Cheat_Sheet.html">https://cheatsheetseries.owasp.org/cheatsheets/Cross_Site_Scripting_Prevention_Cheat_Sheet.html</a> |
| Source Mode | official_catalog                                                                                                                                                                                              |
| Score       | 23.0                                                                                                                                                                                                          |

OWASP recommends context-aware output encoding and layered defensive controls to prevent Cross-Site Scripting. Different output contexts such as HTML body, HTML attributes, JavaScript, CSS, and URL contexts require different encoding rules. Recommended use: Use this reference as the primary remediation guideline for XSS output encoding. Official URL: [https://cheatsheetseries.owasp.org/cheatsheets/Cross\\_Site\\_Scripting\\_Prevention\\_Cheat\\_Sheet.html](https://cheatsheetseries.owasp.org/cheatsheets/Cross_Site_Scripting_Prevention_Cheat_Sheet.html) Source URL: [https://cheatsheetseries.owasp.org/cheatsheets/Cross\\_Site\\_Scripting\\_Prevention\\_Cheat\\_Sheet.html](https://cheatsheetseries.owasp.org/cheatsheets/Cross_Site_Scripting_Prevention_Cheat_Sheet.html)

### 8.2 PortSwigger - PortSwigger Web Security Academy: Reflected XSS

|             |                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source      | <a href="https://portswigger.net/web-security/cross-site-scripting/reflected">https://portswigger.net/web-security/cross-site-scripting/reflected</a> |
| Source Mode | official_catalog                                                                                                                                      |
| Score       | 21.0                                                                                                                                                  |

PortSwigger describes reflected XSS as occurring when an application receives data in an HTTP request and includes that data within the immediate response in an unsafe way. Recommended use: Use this reference to explain reflected XSS behavior and testing methodology. Official URL: <https://portswigger.net/web-security/cross-site-scripting/reflected> Source URL: <https://portswigger.net/web-security/cross-site-scripting/reflected>

### 8.3 CWE - CWE-79: Improper Neutralization of Input During Web Page Generation

|             |                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------|
| Source      | <a href="https://cwe.mitre.org/data/definitions/79.html">https://cwe.mitre.org/data/definitions/79.html</a> |
| Source Mode | official_catalog                                                                                            |
| Score       | 20.7                                                                                                        |

CWE-79 classifies Cross-Site Scripting as improper neutralization of input during web page generation. Recommended use: Use this reference to classify the weakness and map the finding to CWE-79. Official URL: <https://cwe.mitre.org/data/definitions/79.html> Source URL: <https://cwe.mitre.org/data/definitions/79.html>



## 8.4 MDN - MDN Content Security Policy Guide

|             |                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Source      | <a href="https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/CSP">https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/CSP</a> |
| Source Mode | official_catalog                                                                                                                        |
| Score       | 19.7                                                                                                                                    |

MDN describes Content Security Policy as a defense-in-depth mechanism that can help reduce the impact of XSS by controlling script execution sources. Recommended use: Use this reference as a defense-in-depth recommendation, not as a replacement for output encoding. Official URL: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/CSP> Source URL: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/CSP>

# 9. Retest Recommendation

---

Re-run the S2N XSS plugin against the affected endpoint after remediation.

|                        |                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------|
| Target URL             | http://host.docker.internal:4280/vulnerabilities/xss_r/?name=%3Cscript%3Ealert(1)%3C%2Fscript%3E |
| Parameter              | name                                                                                             |
| Payload                | <script>alert(1)</script>                                                                        |
| Expected Secure Result | The payload should be encoded or rejected and must not execute as JavaScript.                    |

## Retest Steps

1. Send the same GET request to the affected endpoint with the original payload.
2. Verify that < and > are encoded as &lt; and &gt;;, or that the input is rejected.
3. Open the response in a browser and confirm that no JavaScript alert or script execution occurs.
4. Re-run the S2N XSS plugin and confirm that the finding count for this endpoint is zero.